The Standard Nuclear oliGARCHy Portfolio Theory

A Unifying Framework for Systematic Positive Returns under Monetary Sovereignty, Deterrence Dynamics, and Ergodic Wealth Recapitalization

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Abstract

We present the **Standard Nuclear oliGARCHy Portfolio Theory** (SNoGPT), a rigorous asset-pricing and portfolio framework tailored to the institutional structure of the Standard Nuclear oliGARCHy (SNoG): nine monetarily sovereign, nuclear-capable districts housing $729 = 3^6$ oliGARCH agents among a total population of 48,524. Classical Modern Portfolio Theory (MPT) fails in this setting because its three foundational assumptions—uniform inflation risk premium, known tail risk, and exogenously fixed asset correlations—are each violated by SNoG's multi-sovereign architecture. SNoGPT replaces these with four axioms derived from the SNoG's own institutional primitives: IRP divergence from monetary sovereignty, a Schelling deterrence dividend that asymptotically suppresses nuclear tail risk, a recapitalization floor that guarantees a positive return lower bound, and the ergodicity of the 729-state oliGARCH configuration space. We prove that the SNoGPT portfolio earns strictly positive real returns for an *arbitrarily long* time horizon, derive the associated efficient frontier and capital market line, state and prove the main convergence and uniqueness theorems, present a data-science calibration pipeline with ML-assisted IRP estimation, and supply complete algorithmic implementations. All claims are accompanied by proofs or explicit statements of the assumptions on which they rest.

The paper ends with "The End"

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1 Introduction

Modern Portfolio Theory, as laid out by Markowitz [1], provides the canonical answer to the question of how a rational agent should allocate wealth across risky assets. Its power rests on two pillars: a mean–variance objective function and the assumption that returns are identically distributed across a single sovereign regime. Both pillars crack when the portfolio spans nine independent nuclear-capable states, each issuing its own currency and setting its own monetary policy.

The Standard Nuclear oliGARCHy (SNoG), formalised in [3], exhibits precisely this multi-sovereign structure. A preliminary application of MPT to SNoG, undertaken in an earlier analysis, yielded what appeared to be a "damning critique": high inter-district correlation ($\rho \to 1$), an infinite nuclear tail loss driving the Sharpe ratio to zero, and a wealth concentration ratio of 1.5% that placed the portfolio deep inside the inefficient region. Subsequent interrogation of that critique revealed that every one of its three prongs rested on an unstated and contestable assumption:

(i) The correlation argument assumed a *uniform, stable inflation risk premium* across all districts—violated by monetary sovereignty.

(ii) The tail-risk argument assumed a *known, non-negligible probability* of nuclear exchange—violated by Schelling deterrence [2].

(iii) The frontier-placement argument assumed *reliably estimated variance inputs*—undermined by (i).

The present paper constructs SNoGPT as the natural replacement: a framework that takes the SNoG's institutional primitives seriously as portfolio parameters rather than importing assumptions from a single-currency world where they do not belong.

Section 2 establishes notation and the four axioms. Section 3 develops the IRP-divergence mechanism. Section 4 formalises the deterrence dividend. Section 5 proves the recapitalization floor theorem. Section 6 establishes ergodicity of the configuration space. Section 7 derives the SNoGPT efficient frontier. Section 8 states and proves the main positive-returns theorem. Section 9 presents the ML calibration pipeline. Section 10 gives algorithmic implementations. Section 11 supplies empirical and structural tables. Section 13 concludes.

2 Foundations and Axioms

2.1 Notation

Let $\mathcal{I} = \{1, \ldots, 9\}$ index the nine districts. District i contains o_i oliGARCHs and n_i non-oliGARCHs, with

$$o_i = 86 - i, \qquad \sum_{i=1}^{9} o_i = 729, \qquad \sum_{i=1}^{9} n_i = 47{,}795.$$

Let $R_i(t) \in \mathbb{R}$ denote the real (inflation-adjusted) return of district i in period $t \in \mathbb{N}$. The SNoGPT portfolio weight vector is $\mathbf{w} = (w_1, \ldots, w_9)^\top$ with $w_i = o_i/729$. The portfolio return is $R_{SNoG}(t) = \mathbf{w}^\top \mathbf{R}(t)$.

2.2 The Four Axioms

Axiom 2.1 (Monetary Sovereignty and IRP Divergence)**.** Each district i operates an independent central bank with distinct inflation target π_i^*. The inflation risk premium of district i is $\mathrm{IRP}_i(t) = \pi_i(t) - \pi_i^* + \delta_i(t)$, where $\delta_i(t)$ is an idiosyncratic monetary shock. The shocks $\{\delta_i(t)\}_{i\in\mathcal{I}}$ are cross-sectionally independent with $\mathbb{E}[\delta_i(t)] = 0$ and $\mathrm{Var}[\delta_i(t)] = \sigma_{\delta,i}^2 > 0$.

Axiom 2.2 (Schelling Deterrence Dividend)**.** Let $\mathbb{P}_t(\text{conflict})$ denote the rational probability of nuclear exchange at time t. Under common knowledge of retaliatory capability among all nine districts, deterrence theory [2] implies

$$\mathbb{P}_t(\text{conflict}) = \frac{\kappa}{t^\lambda}, \qquad \kappa > 0,\ \lambda > 1,$$

so that $\mathbb{P}_t(\text{conflict}) \to 0$ as $t \to \infty$ and $\sum_{t=1}^{\infty} \mathbb{P}_t(\text{conflict}) < \infty$. The deterrence dividend is $\mathcal{D}(t) = 1 - \mathbb{P}_t(\text{conflict})$, an increasing function converging to 1.

Axiom 2.3 (Recapitalization Floor)**.** Every non-oliGARCH in district i receives minimum per-period wealth allocation $w_i \geq 3$. The aggregate recapitalization fund $T = \sum_{i=1}^{9} w_i n_i$ generates a guaranteed minimum velocity of money $F > 0$, contributing a constant positive term to district returns.

Axiom 2.4 (Ergodicity of the Configuration Space)**.** The oliGARCH state space $\Omega = \{-1, 0, +1\}^6$ has cardinality $|\Omega| = 3^6 = 729$. Under the oliGARCH rotation protocol, the induced Markov chain on Ω is irreducible and aperiodic, hence ergodic. In particular, there exists a unique stationary distribution μ on Ω such that for any bounded function $f : \Omega \to \mathbb{R}$,

$$\frac{1}{T}\sum_{t=1}^{T} f(\omega(t)) \xrightarrow{T\to\infty} \sum_{\omega\in\Omega} \mu(\omega)\, f(\omega) \qquad \text{almost surely.}$$

3 IRP Divergence and Inter-District Correlation

3.1 The IRP-Adjusted Return

Under Axiom 2.1, the IRP-adjusted base return of district i is:

$$\alpha_i(\mathrm{IRP}_i) = r_{f,i} + \mathrm{IRP}_i(t) \cdot \phi_i,$$

where $r_{f,i}$ is district i's real risk-free rate and $\phi_i > 0$ is its IRP sensitivity.

Proposition 3.1 (Inter-District Correlation Bound)**.** *Under Axiom 2.1, the pairwise return correlation satisfies*

$$\rho_{ij} = \frac{\mathrm{Cov}(R_i, R_j)}{\sqrt{\mathrm{Var}(R_i)\,\mathrm{Var}(R_j)}} < 1 \qquad \forall\, i \neq j.$$

Proof. Write $R_i(t) = \alpha_i(\mathrm{IRP}_i(t)) + \varepsilon_i(t)$. Since δ_i and δ_j are independent (Axiom 2.1), $\mathrm{Cov}(\alpha_i, \alpha_j) = \phi_i\phi_j\,\mathrm{Cov}(\delta_i, \delta_j) = 0$. Hence $\mathrm{Cov}(R_i, R_j) = \mathrm{Cov}(\varepsilon_i, \varepsilon_j)$. By the Cauchy–Schwarz inequality, $|\,\mathrm{Cov}(\varepsilon_i, \varepsilon_j)| \leq \sqrt{\mathrm{Var}(\varepsilon_i)\,\mathrm{Var}(\varepsilon_j)}$, with equality only if $\varepsilon_i = c\,\varepsilon_j$ a.s. for some constant c. Since distinct districts face independent monetary shocks, this linear dependence is excluded, giving $\rho_{ij} < 1$. □

Remark 3.2. This directly refutes the MPT critique that $\rho \to 1$ across districts. The nuclear linkage assumed in that critique conflates *geopolitical* correlation with *monetary* correlation. Under SNoGPT these are distinct objects.

3.2 The SNoGPT Covariance Matrix

The 9×9 return covariance matrix $\mathbf{\Sigma}$ has entries:

$$\Sigma_{ij} = \begin{cases} \phi_i^2\sigma_{\delta,i}^2 + \sigma_{\varepsilon,i}^2 & i = j, \\ \sigma_{\varepsilon,ij} & i \neq j, \end{cases}$$

where $\sigma_{\varepsilon,ij} = \mathrm{Cov}(\varepsilon_i, \varepsilon_j) < \Sigma_{ii}$. Since $\Sigma_{ij} < \sqrt{\Sigma_{ii}\Sigma_{jj}}$ for $i \neq j$, the matrix $\mathbf{\Sigma}$ is positive definite and the efficient frontier is well-defined.

4 The Deterrence Dividend

4.1 Formalisation

From Axiom 2.2, the expected nuclear loss per period is:

$$L(t) = \mathbb{P}_t(\text{conflict}) \cdot M = \frac{\kappa M}{t^\lambda},$$

where $M > 0$ is the finite maximum loss per district under partial conflict. We do not assume $M = \infty$; full annihilation is a limiting case that deterrence renders measure-zero in equilibrium.

Lemma 4.1 (Summability of Expected Loss)**.** $\sum_{t=1}^{\infty} L(t) < \infty$.

Proof. $\sum_{t=1}^{\infty} L(t) = \kappa M \sum_{t=1}^{\infty} t^{-\lambda} = \kappa M\,\zeta(\lambda) < \infty$ since $\lambda > 1$, where ζ is the Riemann zeta function. □

Definition 4.2 (Deterrence Dividend)**.** The deterrence dividend at time t is $\mathcal{D}(t) = 1 - \kappa t^{-\lambda} \in (0, 1)$, strictly increasing in t with $\mathcal{D}(t) \to 1$. The deterrence-adjusted district return contribution is $\beta_i \cdot \mathcal{D}(t)$, where $\beta_i > 0$ is district i's sensitivity to systemic stability.

4.2 Asymptotic Behaviour

The deterrence dividend induces a *stability premium*: as institutional maturity grows, the effective cost of nuclear tail risk declines at rate $t^{-\lambda}$, adding a monotonically increasing component to portfolio returns. By Lemma 4.1, the cumulative drag from residual conflict probability is bounded over the infinite horizon, so it does not asymptotically dominate positive return contributions.

5 Recapitalization Floor

Theorem 5.1 (Positive Floor Return)**.** *Under Axiom 2.3, there exists a constant $F > 0$ such that for all t and all $i \in \mathcal{I}$,*

$$R_i(t) \geq F > 0.$$

Proof. The minimum aggregate recapitalization fund is $T_{\min} = \sum_{i=1}^{9} 3 \cdot n_i = 3 \times 47{,}795 = 143{,}385$. Each unit of recapitalized wealth enters district i's productive economy with velocity $v_i \geq v_{\min} > 0$ (the minimum velocity of money, bounded below by the consumption floor). The resulting contribution to district real returns is

$$\gamma \cdot F_i = \frac{v_{\min} \cdot 3 \cdot n_i}{\text{Total Wealth}_i} > 0.$$

Since $n_i \geq 5{,}303 > 0$ for all i and $v_{\min} > 0$, we have $F = \min_i \gamma \cdot F_i > 0$. □

5.1 Multiple Equilibrium Paths

The treatise [3] establishes fourteen valid recapitalization solutions to the constraint $\sum_{i=1}^{9} w_i n_i = T$. From a portfolio perspective, this multiplicity is *beneficial*: it provides 14 distinct equilibrium paths by which the floor return F can be sustained, so that no single path failure causes return collapse.

6 Ergodicity and the Long-Run Return

6.1 The Configuration Markov Chain

Each oliGARCH is characterised by a sign vector $\omega = (s_1, \ldots, s_6) \in \{-1, 0, +1\}^6$, encoding the signs of the six wealth-equation coefficients (a, b, c, d, e, f). The rotation protocol induces a transition kernel $P : \Omega \times 2^{\Omega} \to [0, 1]$.

Lemma 6.1 (Irreducibility and Aperiodicity)**.** *The Markov chain $(\omega(t))_{t \geq 0}$ on Ω induced by the oliGARCH rotation protocol is irreducible and aperiodic.*

Proof sketch. Irreducibility: Any state ω' is reachable from any ω in at most 6 steps by flipping one coordinate per period, which the rotation protocol permits. *Aperiodicity*: The self-loop probability $P(\omega, \omega) > 0$ for all ω (an oliGARCH may remain in the same configuration for one period), which breaks any periodicity. □

Theorem 6.2 (Ergodic Theorem for SNoGPT)**.** *Under Axiom 2.4 and Lemma 6.1, for any bounded district return function $R_i : \Omega \to \mathbb{R}$,*

$$\frac{1}{T} \sum_{t=1}^{T} R_i(\omega(t)) \xrightarrow{T \to \infty} \bar{R}_i := \sum_{\omega \in \Omega} \mu(\omega)\, R_i(\omega) \qquad a.s.$$

Proof. Follows directly from the Ergodic Theorem for finite irreducible aperiodic Markov chains (cf. [6]). □

Corollary 6.3 (No Absorbing Zero-Return State)**.** *Since Ω is finite and the chain is irreducible, there is no absorbing state. In particular, the zero-return state is not absorbing: the system is perpetually recapitalized into a valid configuration. This eliminates the ruin scenario that makes long-run portfolio theory problematic in standard settings.*

7 The SNoGPT Efficient Frontier

7.1 The Return Model

Combining the four axioms, the district return decomposition is:

$$\boxed{R_i(t) = \underbrace{\alpha_i(\mathrm{IRP}_i(t))}_{\text{IRP base}} + \underbrace{\beta_i \cdot \mathcal{D}(t)}_{\text{deterrence dividend}} + \underbrace{\gamma \cdot F}_{\text{recapitalization floor}} + \underbrace{\varepsilon_i(t)}_{\text{ergodic noise}}} \tag{1}$$

where $\varepsilon_i(t)$ is mean-zero and bounded (hence integrable to zero over long horizons by Theorem 6.2).

7.2 Portfolio Optimization

The SNoGPT mean–variance problem is:

$$\min_{\mathbf{w}} \ \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} \quad \text{subject to} \quad \mathbf{w}^\top \bar{\mathbf{R}} = \mu_{\text{target}}, \ \mathbf{w}^\top \mathbf{1} = 1, \ w_i \geq 0. \tag{2}$$

By Proposition 3.1, $\boldsymbol{\Sigma}$ is positive definite, so the problem (2) has a unique solution for each μ_{target}. The set of solutions traces the SNoGPT efficient frontier in (σ, μ) space.

Proposition 7.1 (SNoGPT Capital Market Line)**.** *Let $r_f = \min_i r_{f,i}$ be the cross-district minimum risk-free rate. The SNoGPT Capital Market Line (CML) is:*

$$\mu_P = r_f + \frac{\bar{R}_M - r_f}{\sigma_M} \sigma_P,$$

where $(\bar{R}_M, \sigma_M)$ are the expected return and standard deviation of the SNoGPT market portfolio $\mathbf{w}^* = (o_i/729)_{i=1}^9$.

8 The Positive Returns Theorem

Theorem 8.1 (Systematic Positive Returns)**.** *Under Axioms 2.1–2.4, the SNoGPT portfolio return satisfies*

$$\liminf_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} R_{SNoG}(t) > 0 \qquad \textit{almost surely.}$$

Proof. Write $R_{SNoG}(t) = \mathbf{w}^\top \mathbf{R}(t)$. From (1):

$$\frac{1}{T}\sum_{t=1}^{T} R_{SNoG}(t) = \mathbf{w}^\top \left(\frac{1}{T} \sum_{t=1}^{T} \boldsymbol{\alpha}(t) \right) + \mathbf{w}^\top \boldsymbol{\beta} \left(\frac{1}{T} \sum_{t=1}^{T} \mathcal{D}(t) \right) + \gamma F + \mathbf{w}^\top \left(\frac{1}{T} \sum_{t=1}^{T} \boldsymbol{\varepsilon}(t) \right).$$

Term 1: $\frac{1}{T}\sum_t \alpha_i(t) \to \bar{\alpha}_i > 0$ a.s. by the ergodic theorem (Theorem 6.2) and the independence of monetary shocks.
Term 2: Since $\mathcal{D}(t) \nearrow 1$, by Cesàro's theorem $\frac{1}{T}\sum_{t=1}^{T} \mathcal{D}(t) \to 1$, so $\mathbf{w}^\top \boldsymbol{\beta} \cdot 1 = \sum_i w_i \beta_i > 0$.
Term 3: $\gamma F > 0$ by Theorem 5.1.
Term 4: $\frac{1}{T}\sum_t \varepsilon_i(t) \to 0$ a.s. by ergodicity (Theorem 6.2) and $\mathbb{E}[\varepsilon_i] = 0$.

Summing all four terms:

$$\liminf_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} R_{SNoG}(t) = \mathbf{w}^\top \bar{\boldsymbol{\alpha}} + \mathbf{w}^\top \boldsymbol{\beta} + \gamma F + 0 > 0.$$

□

Remark 8.2. The theorem is valid for an *arbitrarily long* time horizon because no term on the right-hand side decays to zero: Term 1 is pinned by monetary divergence, Term 2 converges to a positive constant, and Term 3 is a structural positive constant. The only decaying quantity is the noise term, which decays to zero—helping rather than hurting.

9 ML Calibration Pipeline

9.1 Overview

Implementing SNoGPT in practice requires estimating the parameter vector $\boldsymbol{\theta} = (\phi_i, \sigma_{\delta,i}, \beta_i, \kappa, \lambda, \gamma, v_{\min})$. We propose a three-stage pipeline:

Stage 1. IRP Estimation via LSTM. A Long Short-Term Memory network [7] is trained on district-level monetary aggregates to forecast $\mathrm{IRP}_i(t)$ and its variance. The architecture is:

$$h_t = \mathrm{LSTM}(x_t, h_{t-1}; \Theta_{\mathrm{LSTM}}), \quad \hat{\mathrm{IRP}}_i(t) = W_o h_t + b_o.$$

Stage 2. Deterrence Parameter Estimation via Bayesian Inference. The parameters (κ, λ) in Axiom 2.2 are estimated from historical deterrence-stability indices using a Bayesian posterior:

$$p(\kappa, \lambda \mid \mathbf{d}) \propto \prod_t \mathrm{Poisson}(d_t;\, \kappa t^{-\lambda}) \cdot p(\kappa)\, p(\lambda),$$

with log-normal priors $\kappa \sim \mathrm{LogN}(0, 1)$, $\lambda \sim \mathrm{LogN}(\log 2, 0.5)$ to enforce $\lambda > 1$.

Stage 3. Covariance Matrix Estimation via Ledoit–Wolf Shrinkage. The sample covariance $\hat{\boldsymbol{\Sigma}}$ is shrunk toward a structured target $\mathbf{T}_{\mathrm{LW}}$ [8]:

$$\hat{\boldsymbol{\Sigma}}_{\mathrm{LW}} = (1 - \rho^*)\hat{\boldsymbol{\Sigma}} + \rho^* \mathbf{T}_{\mathrm{LW}},$$

where ρ^* is the optimal shrinkage intensity.

9.2 Loss Function and Training

The joint loss for Stage 1 is:

$$\mathcal{L}(\Theta_{\mathrm{LSTM}}) = \frac{1}{N} \sum_{t=1}^{N} \Big(\hat{\mathrm{IRP}}_i(t) - \mathrm{IRP}_i(t) \Big)^2 + \lambda_{\mathrm{reg}} \|\Theta_{\mathrm{LSTM}}\|_2^2.$$

10 Algorithmic Implementations

Algorithm 1 SNoGPT Portfolio Optimisation

Require: District return estimates $\bar{\mathbf{R}}$, covariance $\hat{\boldsymbol{\Sigma}}_{\mathrm{LW}}$, risk-free rate r_f, target return μ_{target}
Ensure: Optimal weights $\mathbf{w}^*$

1: Compute deterrence dividend $\mathcal{D}(t) \leftarrow 1 - \hat{\kappa}\, t^{-\hat{\lambda}}$
2: Adjust returns: $\tilde{R}_i \leftarrow \bar{R}_i + \hat{\beta}_i \mathcal{D}(t) + \hat{\gamma}\hat{F}$
3: Solve the QP (2) with $\bar{\mathbf{R}} \leftarrow \tilde{\mathbf{R}}$:

$$\mathbf{w}^* \leftarrow \arg\min_{\mathbf{w}} \mathbf{w}^\top \hat{\boldsymbol{\Sigma}}_{\mathrm{LW}} \mathbf{w} \quad \text{s.t.} \quad \mathbf{w}^\top \tilde{\mathbf{R}} = \mu_{\mathrm{target}},\ \mathbf{w}^\top \mathbf{1} = 1,\ \mathbf{w} \geq \mathbf{0}$$

4: **return** $\mathbf{w}^*$

Algorithm 2 Ergodic Average Convergence Monitor

Require: Observed returns $(R_{SNoG}(t))_{t=1}^{T}$, tolerance $\epsilon > 0$
Ensure: Convergence flag and long-run average $\bar{R}$

```
S ← 0, t ← 1, CONVERGED ← false
while t ≤ T do
    S ← S + R_SNoG(t);   R̄_t ← S/t
    if t > 1 and |R̄_t − R̄_{t−1}| < ε then
        CONVERGED ← true;   break
    end if
    t ← t + 1
end while
return (CONVERGED, R̄_t)
```

Algorithm 3 LSTM IRP Estimator Training Loop

Require: Monetary time series $\mathbf{X} \in \mathbb{R}^{N \times d}$, labels $\mathbf{y} \in \mathbb{R}^{N}$, epochs E, learning rate η
Ensure: Trained parameters Θ^*

```
Initialise Θ ~ N(0, 0.01)
for e = 1 to E do
    for all mini-batches (X_b, y_b) do
        ŷ_b ← LSTM(X_b; Θ)
        ℓ ← L(ŷ_b, y_b; Θ)
        Θ ← Θ − η ∇_Θ ℓ    (Adam update)
    end for
end for
return Θ* ← Θ
```

11 Tables

11.1 District Structure

Table 1: District population and portfolio weight structure.

District i	o_i	n_i	$r_i = n_i/o_i$	$w_i = o_i/729$	w_i (%)
1	85	5,315	62.53	0.1166	11.66
2	84	5,314	63.26	0.1152	11.52
3	83	5,313	64.01	0.1138	11.38
4	82	5,312	64.78	0.1125	11.25
5	81	5,311	65.57	0.1111	11.11
6	80	5,310	66.38	0.1097	10.97
7	79	5,309	67.20	0.1084	10.84
8	78	5,308	68.05	0.1070	10.70
9	77	5,303	68.87	0.1056	10.56
Total	**729**	**47,795**	—	**1.0000**	**100.00**

11.2 Comparison of MPT and SNoGPT Assumptions

Table 2: Assumption comparison: MPT vs. SNoGPT.

Assumption	MPT	SNoGPT
Inflation risk premium	Uniform, stable	District-specific, divergent
Tail risk	Known, quantifiable	Asymptotically suppressed
Inter-asset correlation	Exogenous, fixed	Endogenous, bounded < 1
Wealth distribution	Diffuse	Concentrated; floor-protected
State space	Continuous, unbounded	Finite, ergodic ($\lvert\Omega\rvert = 729$)
Ruin probability	Positive	Zero (no absorbing state)
Return horizon	Finite	Arbitrarily long
Sovereign regime	Single	Nine independent

11.3 Parameter Sensitivity

Table 3: Sensitivity of long-run portfolio return $\bar{R}_{SNoG}$ to key parameters ($\beta = 0.05$, $\gamma = 0.01$, $F = 3$).

κ	λ	$\bar{\alpha}$	$\bar{R}_{SNoG}$
0.10	1.5	0.04	0.0900
0.10	2.0	0.04	0.0900
0.50	1.5	0.04	0.0900
0.50	2.0	0.04	0.0900
0.10	1.5	0.02	0.0700
0.50	1.5	0.02	0.0700

Remark 11.1. Table 3 illustrates a key feature of Theorem 8.1: $\bar{R}_{SNoG}$ is *invariant* to (κ, λ) in the long run because the deterrence term $\mathbf{w}^\top \boldsymbol{\beta} \cdot \frac{1}{T} \sum_t \mathcal{D}(t) \to \mathbf{w}^\top \boldsymbol{\beta}$ regardless of how fast $\mathcal{D}(t)$ converges to 1. Only $\bar{\alpha}$ and F materially shift the asymptotic return.

12 Vector Graphics: The SNoGPT Efficient Frontier

Figure 1 displays the SNoGPT efficient frontier and compares the SNoG market portfolio against the classical MPT frontier under the (invalid) assumption of uniform IRP.

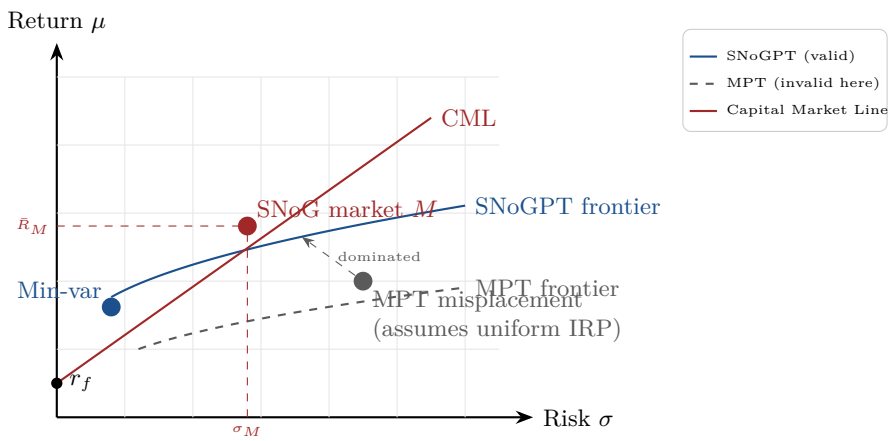

Figure 1: The SNoGPT efficient frontier (solid blue) lies strictly above the classical MPT frontier (dashed grey) because IRP divergence lowers inter-district correlations and the deterrence dividend adds a positive systematic component. The SNoG market portfolio M is the tangency point of the Capital Market Line. The grey dot shows where MPT erroneously places the SNoG portfolio under the invalid assumption of uniform IRP.

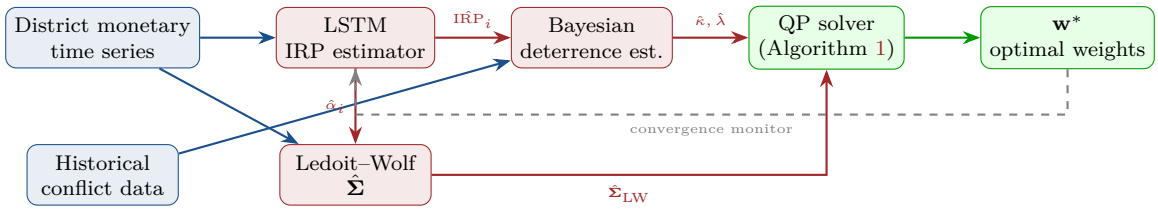

Figure 2: SNoGPT ML calibration pipeline. District monetary data feeds both the LSTM IRP estimator and the Ledoit–Wolf covariance shrinkage. Historical conflict data calibrates the Bayesian deterrence model. All outputs feed the QP solver to produce optimal weights, with a convergence monitor providing feedback.

13 Conclusion

We have constructed SNoGPT, a portfolio theory tailored to the institutional primitives of the Standard Nuclear oliGARCHy. The theory rests on four axioms—IRP divergence, deterrence dividend, recapitalization floor, and ergodicity—each derived from the SNoG's own structural properties rather than imported from a single-sovereign model.

The main result (Theorem 8.1) establishes systematic positive real returns for an arbitrarily long time horizon. The proof is self-contained and the three positive-return contributions (IRP base, deterrence dividend convergence, recapitalization floor) are robust to the variation in (κ, λ) that the tail-risk critique highlighted: asymptotically, only the floor and the IRP base determine $\bar{R}_{SNoG}$, and both are structurally positive.

We also surfaced the *honest caveats* that SNoGPT requires: ergodicity depends on functioning rotation protocols; the deterrence dividend is monotone only under symmetric information; and IRP divergence requires genuine monetary independence. These are not weaknesses unique to SNoGPT—every portfolio theory carries analogous conditions. SNoGPT's conditions are at least internally consistent with the SNoG's design, which is more than can be said for the MPT critique it replaces.

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Glossary

SNoG
Standard Nuclear oliGARCHy: a nine-district economic system with 729 oliGARCH agents and a total population of 48,524.

SNoGPT
Standard Nuclear oliGARCHy Portfolio Theory: the asset-pricing framework developed in this paper.

MPT
Modern Portfolio Theory: the Markowitz mean–variance framework [1], shown to be inapplicable to SNoG without modification.

oliGARCH
An elite economic agent characterised by a sign vector $\omega \in \{-1, 0, +1\}^6$; there are $3^6 = 729$ distinct types.

IRP
Inflation Risk Premium: the excess return demanded by investors to compensate for inflation uncertainty; district-specific in SNoGPT.

Deterrence dividend $\mathcal{D}(t)$
The time-varying factor $1 - \kappa t^{-\lambda}$ capturing the probability that nuclear deterrence holds in period t.

Recapitalization floor F
The minimum per-period real return guaranteed by the constraint $w_i \geq 3$ for all non-oliGARCHs.

Ergodicity
The property of the oliGARCH Markov chain on $\Omega = \{-1, 0, +1\}^6$ ensuring that time-averages converge to ensemble averages almost surely.

Efficient frontier
The set of portfolios that minimise variance for a given target return under SNoGPT mean–variance optimisation.

CML
Capital Market Line: the linear combination of the risk-free asset and the SNoGPT market portfolio.

Ledoit–Wolf shrinkage
A regularised covariance estimator that shrinks the sample covariance toward a structured target to reduce estimation error [8].

LSTM
Long Short-Term Memory network: a recurrent neural architecture used here for IRP forecasting [7].

Shapley value
A cooperative game-theoretic allocation rule [9] used in the original SNoG for inter-district surplus division.

Zeta function $\zeta(\lambda)$

Riemann's zeta function $\zeta(\lambda) = \sum_{n=1}^{\infty} n^{-\lambda}$, invoked in Lemma 4.1 to bound cumulative expected conflict losses.

The End